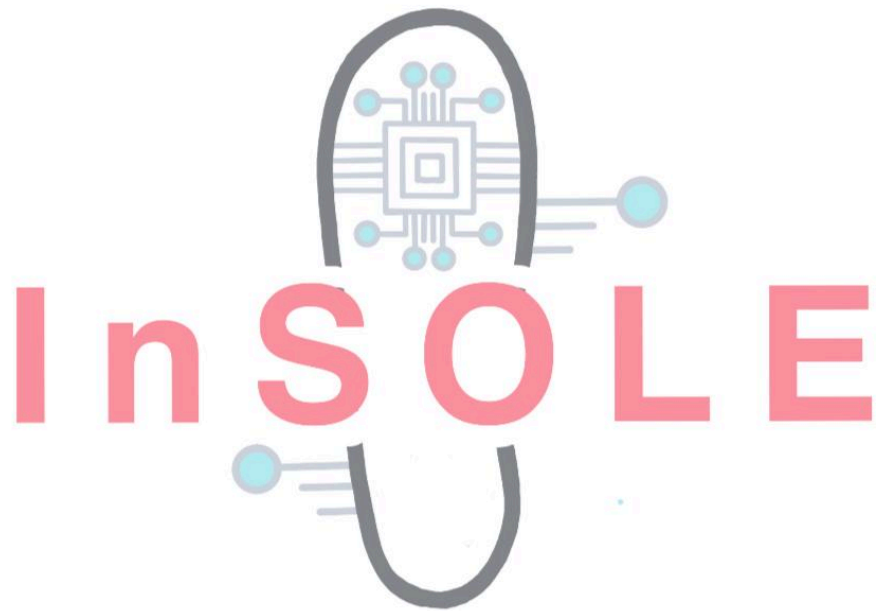




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BME 485 Biomedical Engineering Capstone Design - Final Progress Report

I. Executive Summary

Quantified gait analysis has been proven to play an important role in understanding human locomotion for over 3 decades. However, the application of these systems remains limited to research institutions mainly due to costly equipment such as motion capture labs and force plates [1]. This also creates the need for specialized testing locations. Gait analysis focuses on the mechanics and patterns of human movement in common activities such as walking. Collecting metrics like walking speed and pressure distribution, opens doors for diagnostics, rehabilitation, and enhancing physical performance. There is a need for a portable, cost-effective system that will enable gait analysis to be more accessible, allowing for applications in early detection, monitoring of neurological disorders, and rehabilitation support.

The inSOLE project aims to offer a solution as a high-tech insole with embedded customized pressure sensors. This system will collect multiple data sets to achieve a discrete personalized health monitoring system. The inSOLE strives to bridge the gaps in portability, accessibility, and practicality for individuals that require mobility analysis. This will be done by leveraging modern 3D printing technology, with multifilament printing to develop a system that is effectively manufactured.

As a team we have successfully developed a fully customizable and scalable 3D printed sensor that can seamlessly be encapsulated inside a tailored 3D printed insole. This was completed during the rapid prototyping phase where we were able to fine tune our design and customize it based on subject anatomy. Following this we began to develop our data acquisition protocol. We selected the ideal hardware for our inSOLE and developed a GUI using python tkinter allowing us to capture and export data with an intuitive user-interface. As our last step we completed sensor validation testing and conducted an IRB approved internal human testing.



Figure 1. Conceptual Illustration of inSOLE components

II. Engineering System Design

Objective 1: Research

Task 1.1: Literature review

Our team structure consists of four individuals each possessing different skill sets, offering well rounded knowledge for approaching challenges. During this objective, we collectively reviewed eleven research journals, below is the summarized information that is essential for the advancement of the inSOLE project.

We conducted a literature review on different types of conductive filament. We looked at conductive and non-conductive filaments and narrowed down our search to fit our specific application. We researched PLA (polylactic acid), ABS (Acrylonitrile Butadiene Styrene), PETG (Polyethylene Terephthalate Glycol), and TPU



Figure 2. Serpentine parallel plate capacitive sensor design

(Thermoplastic Polyurethane). Throughout this process, it was important to understand how conductive filaments are made and how the integration of conductive elements affects their properties. We also became familiar with other processes that introduce conductive elements such as surface coating, and blending. [2].

We also revisited the basics of capacitors. While these topics were covered in previous classes, we explored the formulas in greater depth, focusing on understanding each constant. We learned about the dielectric constant which is the measure of how well something can store electrical energy in an electric field. We reviewed how area and distance affects capacitors and how this would relate to our specific project [3] (*figure 2*). Since in our parallel plate sensor distance will vary from person to person due to weight and differing pressures, these basics allowed us to start thinking about ways to create functioning capacitive sensors.

We also researched Force Sensing Resistors (FSRs). These are made up of multiple layers that work in tandem to detect pressure and force. There is a top and bottom flexible substrate layer, a conductive ink layer which creates the conductive traces on the top and bottom, a dot adhesive layer to ensure a controlled contact area, and a piezoresistive layer (conductive polymer). When no pressure is applied the piezoresistive layer, which remains separated from conductive traces, will produce a high resistance. Then as pressure is applied more piezoresistive material touches the conductive traces which lowers the resistance in a nonlinear way meaning larger forces produce smaller incremental resistance changes [4]. This research was important in understanding sensor structure and materials so that we can produce a sensor while fully understanding its properties. However, our sensor will be based on capacitance principles.

The previous work completed in the summer had laid out a strong foundation for our team to continue throughout capstone. Prior to our capstone project there were previous prints of the sensor (*figure 3*). With this foundation we were able to develop our own way to approach the development of the sensor using the previous models and concepts provided. We were also able to gain insights on how to develop the insole through the prototyping process of the summer which investigated different infill patterns, thickness, and densities for an ideal insole (*figure 4*).

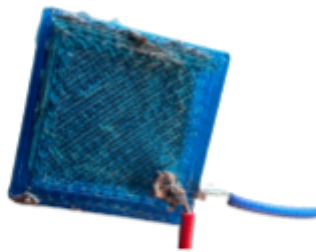


Figure 3: A 3D printed parallel plate capacitive sensor from summer 2024.



Figure 4. Insole prototypes from Summer 2024

Task 1.2: Materials

Through our research we found seven non conductive and conductive filaments that held potential to be used in the project. Our first encountered issue was that we were constricted by the budget. This is due to the conductive filament being far more expensive than generic filaments, but the main characteristics we considered as a deciding factor was the resistivity of each of the filaments. We chose Multi3D Electrifi, NC FilaFlex TPU, and FilaFlex Conductive filament. The conductive Multi3D Electrifi filament is made up of proprietary polymer infused with copper and silver, it also has a resistance of $0.006 \Omega/\text{cm}$. The FilaFlex is a conductive TPU92A filament infused with conductive elements with a resistivity of $3.9 \Omega/\text{cm}$. The FilaFlex is a nonconductive TPU60A.

Our next step was to test these filaments. We modeled and developed a testing protocol to analyze the properties of the filament and understand how it performs when used in the sensor, as seen in figure 3. Another test that was executed in the summer was Scanning Electron Microscopy (SEM), (figure 6). This analysis allowed us to visualize how filaments will react with each other during the printing process. If the filaments do not bond well together, our sensor will not only be inaccurate but also lack durability. This testing gave us a good baseline for how to proceed in our project. It allowed us to tweak how we used and prepared the filaments. Another goal we would like to add is to print the sensor on a textile. This will ensure that the sensor is well adhered and also will provide a "normal" feeling and look to the insole.

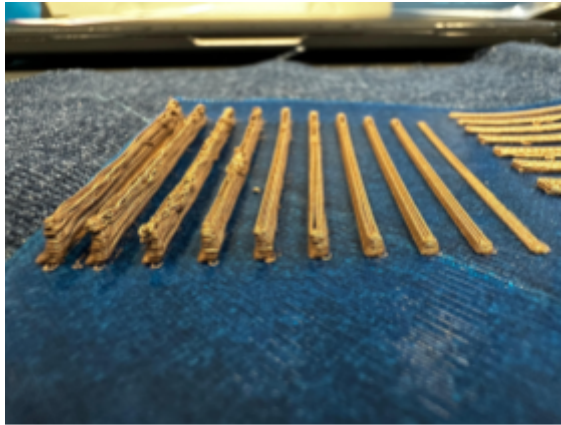


Figure 5. Adhesion and print resolution testing for Multi3D's Electrifi filament done in summer 2024.

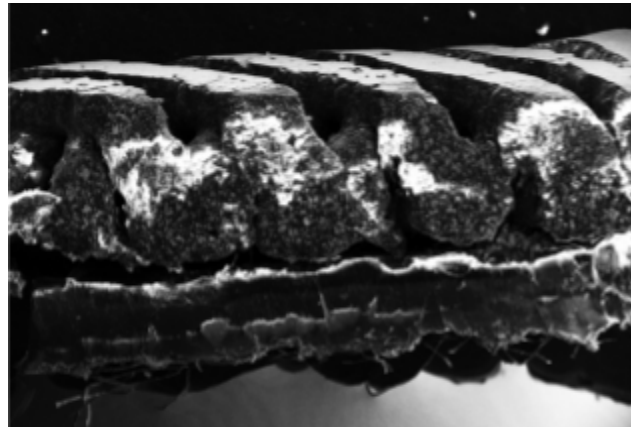


Figure 6. SEM image of summer 2024 sensor. Highlighting the bonding between the conductive layer (top) and the TPU layer (bottom).

Objective 2: Design Process

Task 2.1: CAD & Slicer

The two Design Engineers, Brooke and James, began designing our sensor on Fusion360 and Shapr3D. These softwares allowed them to create multiple prototypes including the parallel plate and lattice sensors. During the design process the thickness and area of the sensors change to allow for best outcome on the Bambu printer while also changing the behaviour of the sensor. After designing the prototypes the STL files were transferred into Bambu Studio to allow for manipulation of the model in the slicer. This slicer converts the CAD models into G-code, which enables us to configure print settings like infill density, layer height, infill pattern, and pauses. This permits us to also change bed temperature, nozzle temperature, and infill speed.

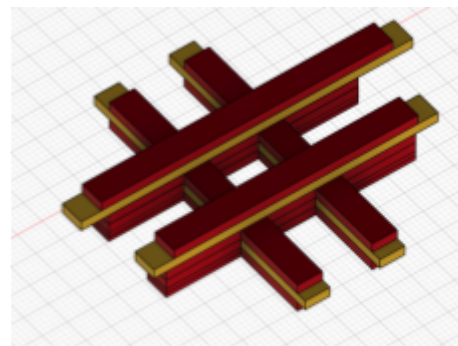


Figure 7. Fusion Model of Capacitive "lattice" sensor design.

Task 2.2: 3D Print

We approached a new sensor structure, deviating from the type that had been developed in the summer. Although the sensor is still designed with the same properties in mind, we started to implement the lattice structure (*figure 7*). We are striving for a sensor that can detect where pressure is being distributed. We also are working on our parallel plate sensor due to simpler functionality and better testing results. This model also allowed easy manual changes in the printer and manipulation of layer heights. After multiple prototypes we transitioned to a serpentine design (*figure 2*). This design allowed for greater sensing area in addition to lowering the surface area to avoid saturation.

Task 2.3: Sensor Development

As we progress through this task and continue to optimize the sensor design, we have printed many variations of the sensor changing shape, layer thickness and filaments to achieve desired operation behavior. Brooke and James took lead in this stage, working to design and print these sensors. The current design features two conductive layers with a NC layer in between (*figure 8*).

Initially we printed these prototypes on the Bambu Lab X1C and P1S printers and encountered compatibility and design challenges. Upon initial print attempts, the print process suffered from frequent jamming due to compatibility issues with TPU in the bambu printer. These issues could be resolved by tweaking printing parameters. We also discovered issues regarding the purging of filament in between manually switching filaments, as the first prototypes of the NC layer contained traces of conductive filament. This would create undesired qualities in the sensor. Due to the issues experienced with the FilaFlex 60A we switched to generic 95A TPU as this material is compatible with most printer configurations. As we gained more prototypes, we further changed the thickness of the sensors and overall shape. The circular design printed exceptionally as this shape reduced the amount of sharp changes in the extruder, resulting in less stringing and an overall quality print.



Figure 8: These are the prototypes in order from oldest to newest.

The printing settings we found most helpful were infill density, infill speed, and nozzle temperature. We decreased the recommended infill speed to 20mm/s and increased infill density to 100% to ensure conductivity. We increased the nozzle temperatures for both FilaFlex 92A and Generic 95A to above their recommended values due to the Bambu printer not allowing TPU to be utilized in the AMS. Manually loading and unloading the filaments mean temperatures needed to be increased for best adhesion. We started with layer thicknesses ranging from 0.4-1mm. Our initial print was manufactured without added supports to the arms so we could see the flaws that developed throughout the print, highlighting the locations that needed structural aid. Our last sensor ended up with 1mm for NC and 0.8mm for the conductive plates. These dimensions were chosen to alter the sensitivity of the sensor.

After encapsulation testing we found that FilaFlex92A was not conductive enough to account for desired sensor behavior. Following



Fig. 8: Serpentine design integrated into inSOLE system

this discovery we switched to Multi3D Electrifi Conductive Filament for its high conductivity and printability. For seamless sensor integration in the printing process we transferred over to the Raise3D Pro2, which featured dual extruders. With this printing setup we integrated our sensors into the insole design. After conduction of testing we found the need for increased sensing area, which the serpentine design offered (*figure 3*). During prototyping with this sensor design we experimented with changing the dielectric layer thickness and conductive plate width to optimize charge and discharge flow (*figure 9*)



Fig. 9: Prototyping of sensors encapsulated in the insole.

Task 2.4: Hardware

After the development of the insole we can move onto hardware. Bella will take the lead on this. The hardware consisted of a 3.7 Volt battery, two ESP32 S2 Thing Plus microcontrollers, a FDC1004 front end capacitance to digital converter. The battery, ESP-32 transmitter, and FDC1004 front end chip were all enclosed in a 3D printed box which we secured to the top of a shoe via the shoelace. This 3D printed box has compartments for each of these three components for a seamless integration.

A 3.7 Volt battery is used for its compact nature as well as its ability to stay within the safe voltage ranges of the microcontroller. The ESP32 S2 Thing Plus is used for its wireless wifi communication protocol known as ESPNOW. This allows the transmitter microcontroller to be connected to our insole and a receiver connected to a computer collecting real-time data. We use the 4-channel capacitance-to-digital converter, the FDC1004 front end chip for its capacitive sensing capabilities and high resolution as well as its ability to easily communicate with the ESP32. [5] We are using environmental monitoring so we can indicate where environmental conditions create changes in capacitance.

The 3.7 Volt battery powers our microcontroller transmitter which uses I2C connections to communicate with the ESP. Which is then soldered into the FDC chip. We connected wires to the Power Processing Unit (PPU) and inserted them into the exposed Multi3D Electrifi filament of the two sensors and ground. Channel 1 and channel 2 on the front end chip are connected to the heel and toe sensors respectfully.

Task 2.5: Coding/Software

Data acquisition first began by using the arduino IDE to code our transmitter and receiver ESP32s. This allowed us to designate a specific sampling rate of 22 Hz. This was due to delays we needed to add in order to not lose data. The transmitter code was used to configure, read, calibrate, and send the data every 45 ms. This code also utilizes threading to receive data simultaneously from two sensors. The receiver code listens for the wireless data from a transmitter ESP32 and displays the data on a computer. The upload of each of these codes is only required once, after which data can be streamed continuously.

The receiver arduino code sends serial output to Python. Now that the code is now streaming into the Python, peak detection and thresholds can be used for desired information. In Python a 3-window graphical user interface (GUI) was designed consisting of a self-identification, data collection, and

analysis screen (*figure 9*).

The self-identification screen is where a user will be asked gender assigned at birth and height. This information will be stored away for later and compared to reference data when the collected data gets analyzed. Once the user presses submit the second screen will appear and data collection will appear.

The data collection first begins by selecting the port that your ESP32 receiver microcontroller is connected to. The user will then proceed to press the start button which will start the data collection and can be seen in the real-time data plot. The data on the plot is on a running time and auto scales to the maximum and minimum values shown on the plot for easier observation. After a few steps are completed, the user will be instructed to press auto threshold, this creates an independent threshold for each sensor that is 40% above each sensor's minimum value. This is what sparks the calculations of the raw data. It is important to code auto thresholds before the calculations begin as you will be provided with outlier data and this will skew the analysis. Once the auto threshold is pressed the code will only recognize the peak that reaches the threshold on the way up. The code is told to calculate step count, stride time, swing time, and stance time. Step count is calculated when it recognizes a heel-strike. Stride time represents a full gait cycle so this is the measure of time between two heel strikes, so the code takes a user's current heel strike minus a user's previous heel strike. Swing time is the amount of time a user's foot is in the air so this is measured by finding the time between current heel strike and previous toe off. (current heel strike - previous toe off). Stance time is the time your foot is on the ground, this is the measured time between previous heel strike and previous toe time. (previous toe off - previous heel strike). Once enough data is recorded, the user will press stop then analyze and this will move the third screen where the gait analysis transpires. This data can also be saved to a csv using the save data button for further analysis and inspection.

The analysis screen first begins by telling the user their cadence. This is calculated average steps per minute. The analysis then shows average stride time. This averages all the used stride time collected in the previous screen. Based on reference data found from a journal we were able to set a classification within the code for a person's gender and height correlated to average stride time. [6] This limited us to men who ranged from 4.7 ft to 6.3 ft in height and women who ranged from 4.9 ft to 5.9 ft in height. This allowed the user to see if their steps took longer or shorter than normal. The analysis screen also showed the average stance and swing time which again just took all the stance and swing times collected in the previous screen and averaged them. To consider if a user has normal gait their cycle needs to consist of $40\% \pm 5$ swing time and $60\% \pm 5$ stance time. Percent stance was calculated by taking average stance over average stride time and multiplying by 100. Percent swing was calculated by taking average swing over average stride time and multiplying by 100. Based on the users results it would show whether the calculated gait was normal or abnormal. This GUI allows for a more user friendly experience and allows us to demonstrate the product with ease.



Figure 10: Here are the 3 GUI screens presented to the user.

Objective 3: Testing

Task 3.1: Data Collection

After completion of software design we transitioned to collecting user data and validating our insole. As our Systems Engineer, Jack performs these tests, relaying the findings to the team before the project progresses to further stages. As a team we received IRB approval to run external testing, but due to time constraints we were only able to perform internal testing. Once a user was fit for their personalized insole, our team would start the python code. We would instruct the participants to take 3 steps then after calibration we would have the participant take 10 steps. We would then analyze and review results. We conducted 10 trials in the same manner.

Engineering Standards for Medical Devices

There are several IEC and ISO standards that may be applicable to this design. The first is IEC 60601-1-2 : Medical Device Electromagnetic Compatibility. We will need to test how our system interacts with electrical components around it. One notable interaction that comes to mind is that with an internet connection. Our interface with a microcontroller is dependent on wireless internet. Another standard would be IEC 60601-1-6 Medical Device Usability. This standard covers validating the personal usability of a product. In our project, we foresee the need to personalize any insole we create for specific shoe sizes, which will require validation for individual compatibility alongside the basic functionality of the sensor. A final and very important standard to consider would be ISO 14971: Risk Management. Within this standard, we have to consider the risk of harm and incident, and the potential severity. At this stage in our process, risks such as comfortability emerge, as an uncomfortable design may lead to balance issues or movement related injury.

FDA and Medical Device regulations

Our device could classify as either a FDA class 1 or Class 2 device due to its low risk. Depending on the direct application of our device, it may not even be necessary to register under the FDA as a medical device. If our device's usage is meant for general human wellness, it does not need to be registered. If it is being used for diagnostic purposes, which our user interface and analysis system does to some extent, it will need to be registered. Manufacturing protocols such as GMP (Good Manufacturing Processes) under a QSR (Quality Systems Regulation) would also be necessary, as our manufacturing process is just as vital as the end product.

III. Testing and Results

We validated the toe and heel sensors to assess that any data gathered would be accurate and precise. The two major tests we chose to conduct were hysteresis and a gauge factor calculation. We chose to measure the capacitance changes over an increase and decrease of load at specific intervals of time. We deemed hysteresis to be an important factor to measure because our sensor is expected to handle dynamic changes in force, and our system depends on there being an accurate data peak for each heel or toe strike. Gauge factor is important, because it is a good estimation about how well a sensor responds to different load types. We would expect our hysteresis traces to be as consistent in amplitude as possible, and our gauge factors to be linear and increasing. To conduct the hysteresis test, we used a timer and SerialPlot software, increasing the weight by one hundred grams until we reached five hundred grams every 10 seconds. We then decreased the load back to zero in the same increment and time span. We created a graph of the forward and backward trace using OriginPro graphing software. We found the heel sensor to be 98.2% , while the toe sensor was 97.4% consistent. We calculated our gauge factor with a forward progression, taking the average capacitance of the whole sequence. Again, our heel sensor was found to be stronger, with its gauge factor having an entirely positive and higher range than the toe sensor.

We validated our user interface system's analytics by visual inspection and by utilizing a multimeter. Our interface system calculates statistics such as stance and swing time for each individual step. We would average the individual steps, then compare to the average printed at the results screen. To validate our step counter system, we did ten internal trials. In each trial, we would have the user take three calibration steps to evaluate the capacitance threshold at peak, then take ten working steps. Within our ten trials, ninety three out of the one hundred total steps were recorded. We feel satisfied with our results for this prototype, but recognize that more can be done to reduce error.

IV. Project Management and Timeline

James Cook (Team Leader and Design Engineer II):

As the team manager, James plays a key role in facilitating the planning and organization of the group. He also takes the lead in the integration and design of the insole, ensuring that all components work seamlessly together. As a team manager he has the responsibility to help the group form a productive way to communicate questions or updates. James establishes clear goals for the week and has check-ins with each group member to receive progress reports. He also helps to keep the group on track with our schedule and communicate with Capstone leaders if assistance is needed. He also aids in the design process with Brooke Costin. Both James and Brooke create prototypes and utilize the 3D printer to print them out.

Brooke Costin (Design Engineer I) -

As Design Engineer Brooke Costin leads sensor design and prototyping. She collaborates with James to develop the project's design using various tools and techniques. The team began by training on the 3D printer with Matt to understand its functionality, starting with 3D printing a "benchy" as practice. Once familiar with the equipment, she initiated the design process, taking responsibility for creating the initial design. Additionally, she creates the CAD model, ensuring the sensor integrates seamlessly with the other components of the project by working closely with the rest of the team. She adjusts the sensor design based on group feedback and manages both the 3D printing tasks and the materials, accounting for any design challenges that arise.

Bella Nealy (Software Engineer) -

As a Software Engineer Bella Nealy leads in the data acquisition system and GUI. This means she is responsible for developing the code for our embedded system in our insole. She is using C++ and

Python to create a data acquisition system for our insole. She also is responsible for constantly changing the code in terms of what we will need. As Well as debugging the code and making sure it integrates well with the other components in our project. She also utilizes the ESP32 microcontroller, which features built-in Wi-Fi capabilities, enabling seamless one-way wireless communication. This functionality is essential for our insoles, allowing them to effectively transmit data to our computers and receive instructions or updates in return. She is also working with the FDC1004 capacitive-to-digital converter in order to obtain capacitive sensing.

Jack Vieira (Systems Engineer) -

As a Systems Engineer Jack Vieira leads closed testing and human testing along with analysis. He works closely alongside Bella to come up with a way to collect data for the insole in both these tests. He comes up with ways to test our project and work with Bella to come up with a way to collect data for the testing. He is responsible for testing the resistivity and capacitance of all the printed sensors to ensure the prints meet the required specifications and functionality. Once the insole is ready for testing, it will be printed to his specifications and modified based on the information collected from these tests. Once we are ready for human trials, he will be responsible for reaching out to the track team and analyzing results.

V. Conclusions & Future Work

The InSOLE project aimed to design, manufacture, and validate the 3D printed capacitive sensor's abilities to be used for gait analysis capabilities. Through months of discovery and testing, the team developed a printing method that allowed for a functional sensor with high accuracy and sensitivity, as well as a data collection, storing, and analysis system with relevance to real world applications. Through multiple testing criteria, the sensors were found to be fully responsive to the human gait pattern. The team members overcame challenges such as material inefficiencies and printing drawbacks. Overcoming the printing liabilities took innovation, days of redesigning, and careful and constant oversight of the prints occurring. Further work can be done to include bilateral testing, as well as a more compact and unobtrusive model.

VI. ABET Outcomes

1:	An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics During the material testing and sensor design phases of this project, the team is tasked with designing a sensor that can handle specific amounts of force produced by day to day motion of the human body. We use fundamental equations from principles of electrical engineering to create these sensors. We also use knowledge of material science to choose our filaments, to ensure that the filaments react and adhere to each other when layered.
2:	An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors. Within our sensor design and production, we want to keep in mind that with a wearable, the comfort of the wearer needs to be addressed. A comfortable sensor will be the safest sensor for the wearer, as it should provide an unobtrusive experience. Our insole design will need to be specifically tailored to the user, ensuring personal comfort and precision of data. At the end of

	our tenure on the project, we created an apparatus threaded through the laces of the attached sneaker. We did this to keep the microcontroller out of the direct footpath, keeping the user's feet from stepping on or coming into contact with the device.
3:	An ability to communicate effectively with a range of audiences Our group has met with a variety of faculty and peers using multiple communication methods. We have communicated in person, on Zoom meetings, and through email and texts. Our end of semester presentation has been organized and rehearsed to be cohesive and on level for a variety of audiences, ranging from below college students to our professors.
4:	An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts. In the testing phase of our project, we must follow the ethical and professional guidelines set before us. As we are doing human testing, we may need IRB approval. Regardless of the need for confirmation, we will obey any and all requirements preordained. We received IRB approval prior to our in house testing, each member reading and discussing the information needed in this phase.
5:	An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives. Our team has specific roles that often intertwine depending on which tasks are being done. Working together to communicate and complete the tasks increases performance and individual efficacy. For example, the Systems Engineer and Software Engineer will work together during the data collection phase.

VII. References

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Thank you, Kunal Mankodiya, Dhaval Solanki, and Suparna Veeturi for your help as Capstone advisors this semester. We appreciate your support in coordinating this program

Appendix A

Updated Timeline: [inSOLE Gantt Chart](#)

Meeting Notes:

Meetings were scheduled on Monday mornings before capstone class, we established with Rumon the topics we would like to discuss before hand and would estimate meeting duration based off this. Multiple master plans were made in a gantt chart manner to ensure productivity and that tasks were being carried out in a timely fashion. We prepped a multitude of questions for our mentor that would address current challenges we were facing on a week to week basis.

Additional meetings could be scheduled based on availability, upcoming deadlines, and arising issues in the project.

We also were able to communicate via text and email with Rumon during times of scheduling conflicts, illness, or other unforeseen circumstances.

Appendix B

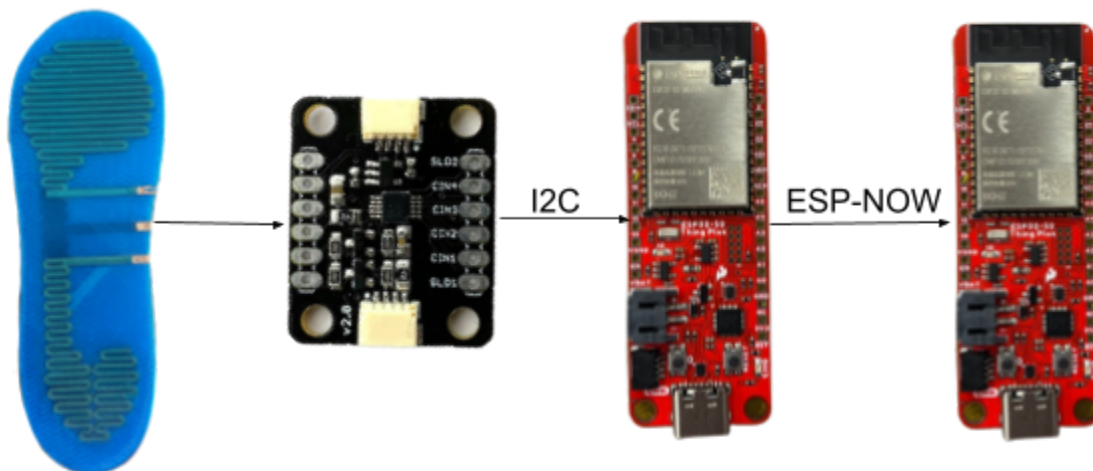
Hardware documentation including schematics, CAD files, bill of materials, etc.

Insole Bill of Materials:

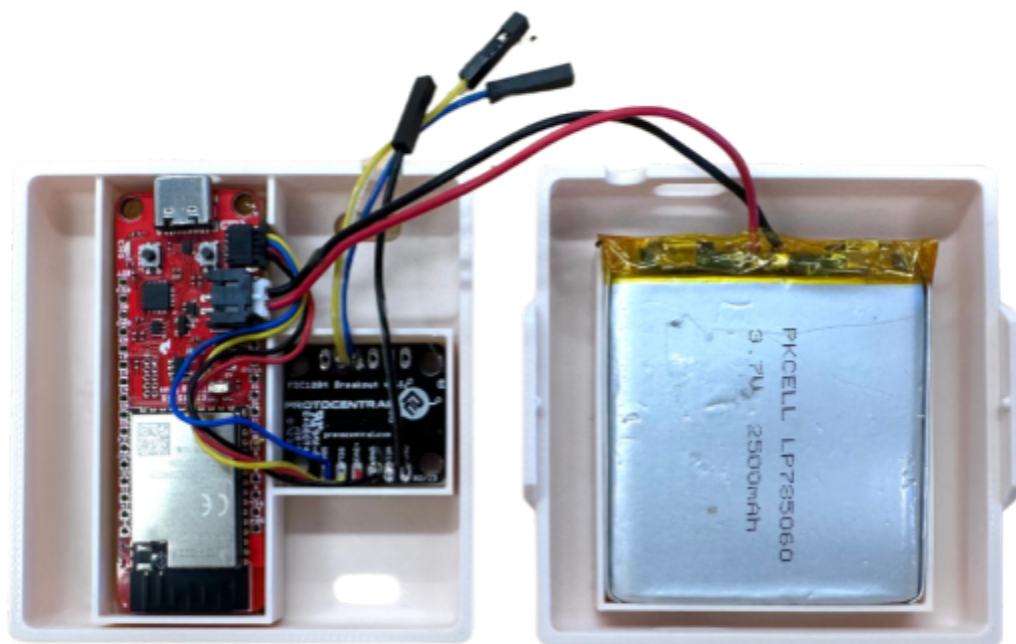
Bill of Insole Materials					
Filament	Vendor	Quantity	Volume	Price	Link
FilaFlex TPU	Recreus	2	500 gm	\$57.90	https://recreus.com/gb/filaments/1-1-filaflex-60a.html#/1-colour-black/2-diameter-175 mm/3-weight-500 gr
FilaFlex Conductive	Recreus	1	500 gm	\$69.96	https://www.3djake.com/recreus/conductive-filaflex-black
Electifi	Multi3D	1	100g	\$215	https://www.multi3dlc.com/product/electrifi/

Hardware Bill of Materials:

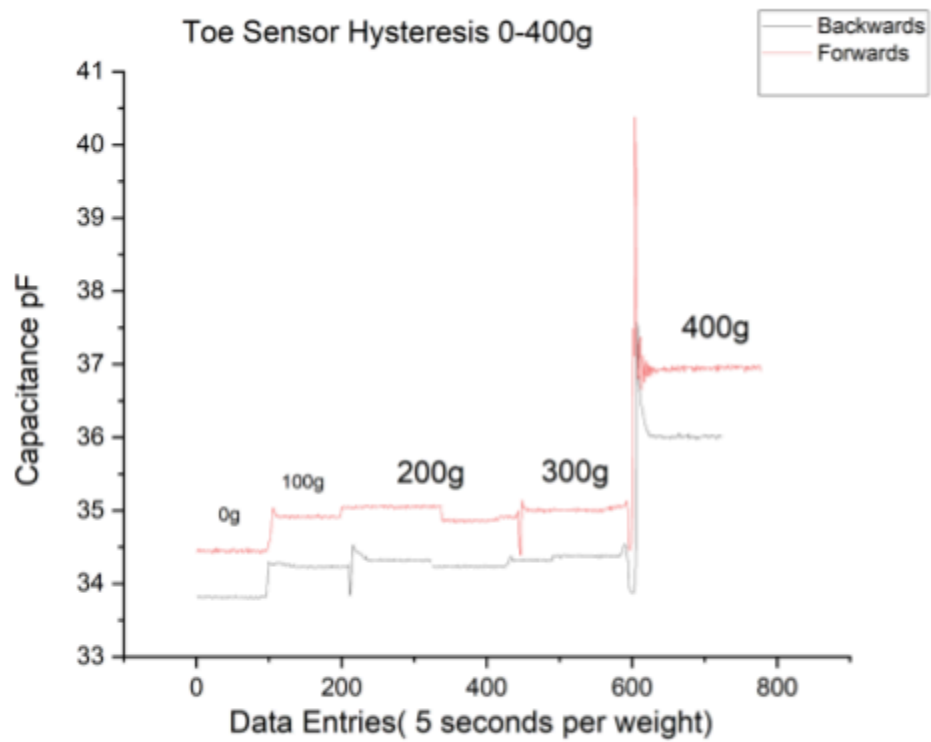
Bill of Hardware Materials					
Hardware	Vendor	Quantity	Price/unit	Total	Link
ESP32 S2 Thing Plus	Sparkfun Electronics	2	\$24.50	\$49	SparkFun ESP32
4 Channel FDC1004 front end chip	Protocentral	1	\$19.91	\$19.91	Protocentral FDC1004
3.7 Volt Battery	Adafruit	1	\$7.95	\$7.95	Adafruit 3.7 Volt Li Battery



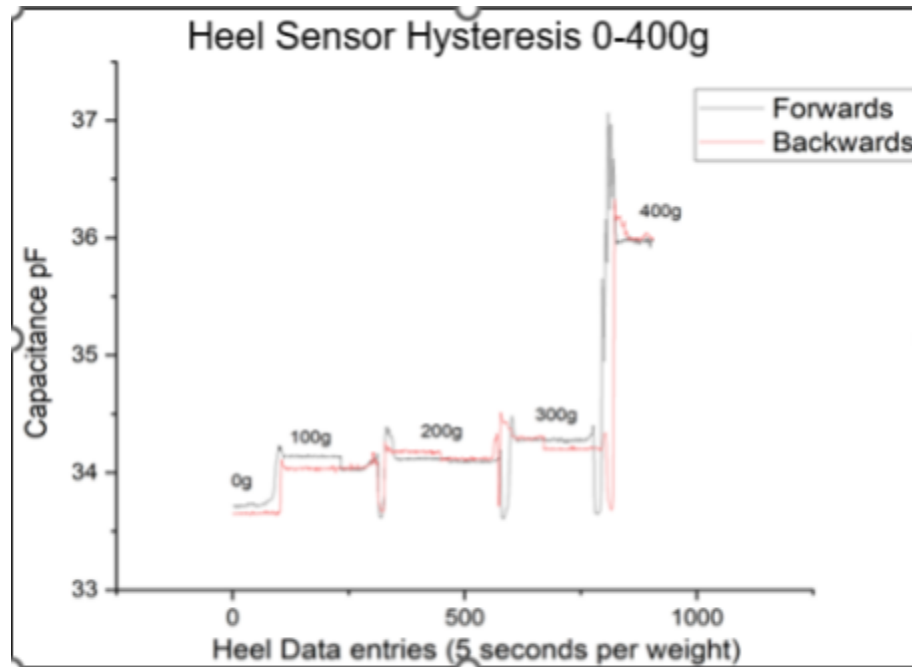
Hardware Schematic



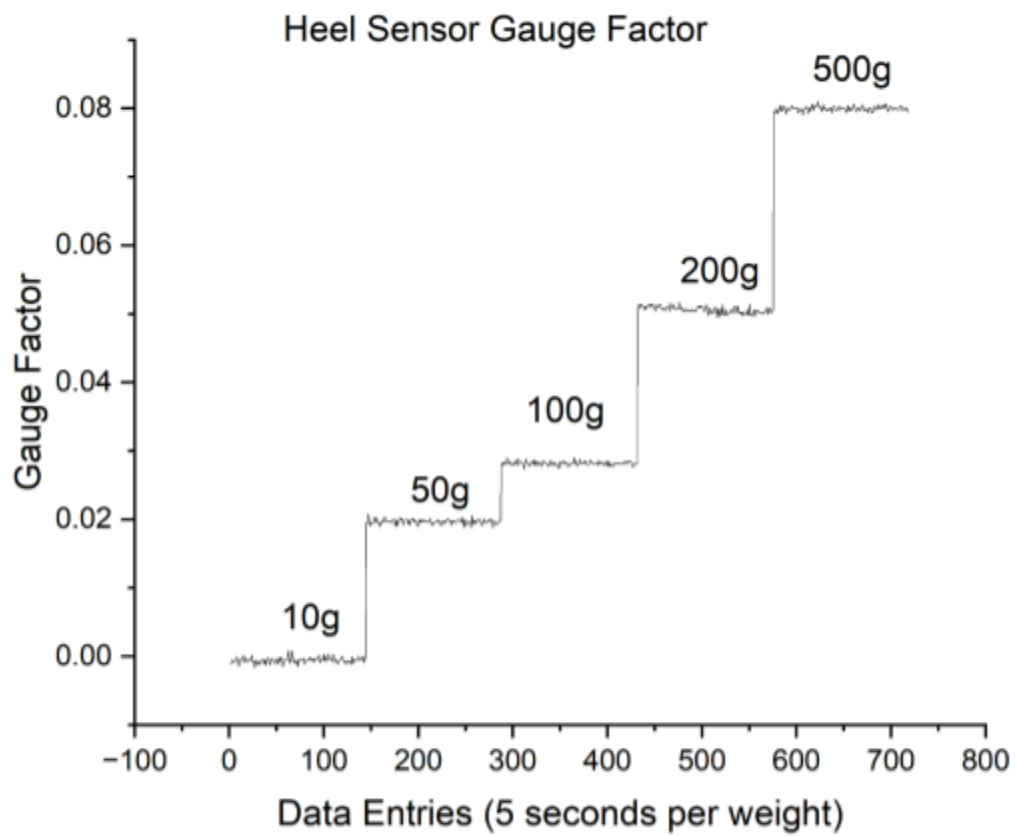
Power processing unit in individual compartments within 3D printed enclosure.



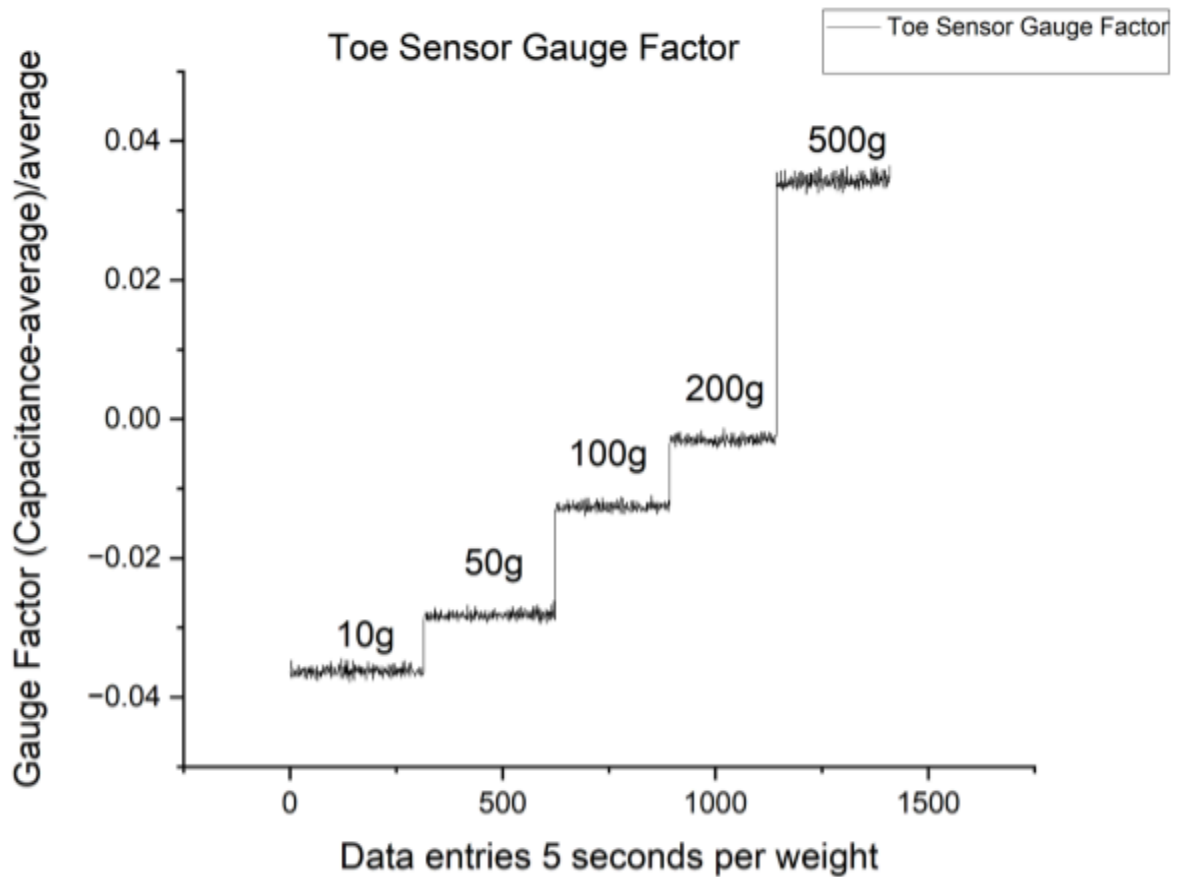
Toe Sensor Hysteresis Data



Heel Sensor Hysteresis



Heel Sensor Gauge Factor



All graphics, including these can be found in our Google Drive folder

▣ Origin Graphics

Appendix C

Software documentation including the list for source code(s).

[Arduino Transmitter Code](#)

[Arduino Receiver Code](#)

[Python GUI](#)

[Insole Trials \(Recorded Steps\)](#)